

Received 28 November 2025, accepted 15 December 2025, date of publication 25 December 2025, date of current version 5 January 2026.

Digital Object Identifier 10.1109/ACCESS.2025.3648539

TOPICAL REVIEW

AI-Enabled Vehicular Data Offloading for Sustainable Smart Cities: Taxonomy, KPI Models, and Open Challenges

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The authors extend their appreciation to Prince Sattam bin Abdulaziz University for funding this research work through the project number (PSAU/2025/01/35351).

ABSTRACT Vehicular edge computing (VEC) is emerging as a key enabler for intelligent transportation systems that are both latency- and energy-sensitive. This survey is motivated by the need for a unified, KPI-driven view of AI-based vehicular computation offloading that explicitly links performance gains to sustainability objectives in smart cities. We synthesize recent advances in AI-powered offloading for vehicular networks, with emphasis on deep reinforcement learning (DRL) and multi-agent variants that learn adaptive, sequential policies under dynamic topology, fluctuating wireless capacity, and heterogeneous workloads. We propose a unified taxonomy that spans infrastructure-based, vehicle-assisted, and hybrid architectures; map offloading decisions to key performance dimensions (end-to-end latency, energy efficiency, reliability, throughput, and task-success rate); and formalize a minimal KPI model that links radio, compute, and caching components. The review compares algorithmic designs (DQN/DDPG/A3C/SAC, prioritized and federated variants, DRL+optimizer hybrids), scheduling granularities and baseline choices, while examining reproducibility factors such as simulators, mobility models, and dataset availability. We further discuss integration with enabling technologies (cellular vehicle-to-everything (C-V2X)/NR-V2X, reconfigurable intelligent surfaces (RIS), UAV relays, edge caching), security and privacy considerations, and the sustainability implications of AI-driven offloading for intelligent urban environments. The paper concludes with open challenges including non-stationarity, sim-to-real transfer, safety constraints, and explainability and outlines a research agenda toward robust, accountable, and resource-efficient offloading policies deployable in real world VEC systems.

INDEX TERMS Vehicular edge computing (VEC), computation offloading, artificial intelligence, edge caching, sustainability, smart cities, SDG11, SDG13, SDG7, SDG9.

I. INTRODUCTION

The rapid increase in data traffic in urban vehicular networks is attributed to the growing prevalence of connected and autonomous vehicles, supported by advancing smart city infrastructures. Applications such as autonomous driving, real-time navigation, traffic monitoring, and high-resolution

infotainment produce significant data volumes that require efficient energy use, high reliability, and low latency. The increase imposes a significant strain on the existing cellular infrastructure, resulting in congestion and deterioration of Quality of Service (QoS), especially in densely populated urban regions [1], [2], [3], [4], [5], [6], [7]. Inefficient data management directly impacts sustainability objectives by increasing energy consumption, carbon emissions, and urban congestion, thus undermining the objectives of SDG 11

The associate editor coordinating the review of this manuscript and approving it for publication was Jiankang Zhang^{ID}.

(Sustainable Cities and Communities) and SDG 13 (Climate Action).

A viable solution has arisen in data offloading, enabling vehicles to transmit data via alternative communication technologies such as Wi-Fi, roadside units (RSU), and vehicle-to-vehicle (V2V) or device-to-device (D2D) connections. This approach strengthens the resilience of urban mobility systems and minimizes greenhouse gas emissions by optimizing latency, enhancing energy efficiency, and ensuring uninterrupted service continuity, thus easing the strain on cellular networks. Conventional offloading strategies often struggle to adapt to the dynamic nature of vehicular environments, which are marked by unpredictable topology changes, varying channel conditions, and shifting service demands [8], [9], [10], [11], [12], [13], [14]. Inability to adapt can result in poor resource utilization and increased operating energy demands, which undermines sustainability initiatives.

Recent advances in Artificial Intelligence (AI), such as deep learning, machine learning, and deep reinforcement learning (DRL), present considerable opportunities to improve data-offloading systems in smart sustainable cities. Deep Reinforcement Learning (DRL) methodologies, including algorithms such as Deep Q-Networks (DQN), Double DQN (DDQN), and policy gradient techniques such as Proximal Policy Optimization (PPO), have been used effectively to improve task and data offloading in vehicular edge computing [2], [15], [16], [17], [18], [19], [20]. LSTM-PPO models demonstrate the capability to accurately predict the load of the base-station (BS) and minimize processing delays during periods of congestion [1], [5]. Multi-agent DRL frameworks enable collaborative offloading and resource allocation [14], [20], [21], thus enhancing energy efficiency and promoting environmentally sustainable network operations.

In addition, hybrid AI strategies that combine supervised learning, reinforcement learning, and optimization techniques have shown significant decreases in energy consumption and latency in VANETs [22], [23]. Recent research highlights the importance of AI in aligning vehicular data management with the sustainability goals of smart cities, including improving urban resilience, decreasing emissions, and improving infrastructure efficiency, along with algorithmic performance [24], [25]. Despite advances, the field remains fragmented, characterized by a lack of unified frameworks or reproducibility benchmarks, along with diverse methodologies, AI models, performance metrics, and network contexts [10], [26], [27]. Figure 1 presents the overall workflow of AI-driven offloading in urban vehicular networks, highlighting the interactions between vehicles, RSUs, and edge / cloud layers.

Motivation: The primary motivation behind this survey is to provide a unified, KPI-driven view of AI-based vehicular data offloading in vehicular edge computing. Existing surveys typically focus either on individual technologies (e.g., VANETs, MEC, or specific AI methods) or on

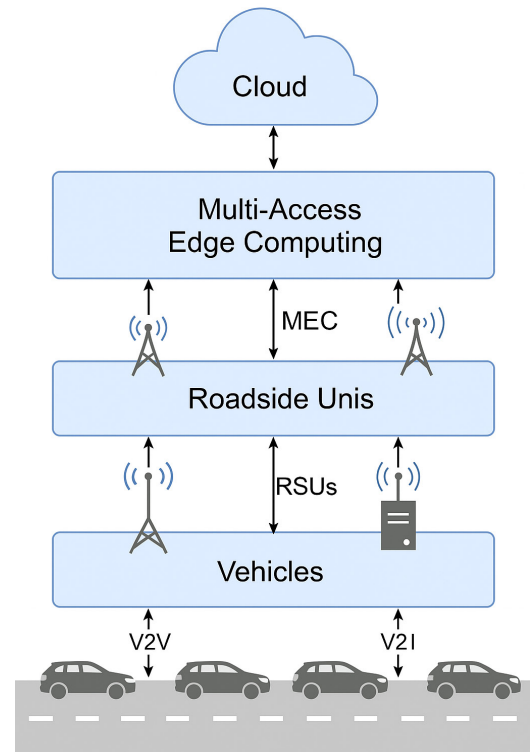


FIGURE 1. Overview of AI-enabled data-offloading workflow in urban vehicular networks.

high-level V2X architectures, but they rarely: (i) compare classical machine learning, deep learning, deep reinforcement learning, federated/distributed learning, and meta-/transfer-learning within a common taxonomy; (ii) relate their reported gains to a consistent set of latency, throughput, energy, and reliability metrics; and (iii) explicitly connect these gains to sustainability and SDG-oriented objectives for smart cities.

This survey is therefore intended to synthesize a fragmented literature into a coherent framework that helps researchers and practitioners select appropriate AI paradigms for vehicular offloading under concrete KPI targets and sustainability constraints.

A. CONTRIBUTIONS

Several comprehensive surveys on vehicular offloading and vehicular edge computing (VEC) exist, spanning network architectures, offloading strategies, and orchestration (e.g., [17], [33], [34], [35]). Table 1 contrasts these reviews along four axes central to this work: explicit sustainability/SDG framing, a unified latency–energy KPI model, integration of emerging technologies (blockchain, digital twins, RIS, SDN, carbon-aware scheduling) and practices for verification and reproducibility.

As shown, previous surveys typically emphasize architectures and algorithms, but stop short of linking offloading mechanisms to environmental outcomes or of standardizing KPIs across radio, compute, and caching layers.

TABLE 1. Comparative positioning of this survey relative to existing reviews on vehicular/edge offloading.

Study (year)	Methodological lens	Latency/energy modeling	model-	Sustainability/SDG linkage	Emerging technologies (BC/DT/RIS/SDN)
Liu et al. (2019) [28]	Systems/VEC orchestration; Vehicle-to-Infrastructure (V2I), Vehicle-to-Vehicle (V2V), Vehicle-to-Everything (V2X) taxonomy	Latency and QoS metrics; energy discussed qualitatively; cross-layer view		Efficiency framing; operational considerations	SDN and MEC; early edge-integration
Zhou et al. (2018) [29]	Network-centric offloading; incentives and pricing in VANET/V2X	Latency/throughput formulations; radio-link energy references		Technical performance emphasis	SDN references within vehicular offloading
Islam et al. (2021) [30]	Multi-access Edge Computing (MEC) task-offloading taxonomy; decision variables and constraints	Optimization models; heterogeneous latency/energy metrics across layers		Future-oriented sustainability notes	SDN examples; ML/FL mentions in MEC
Lin et al. (2020) [31]	Modeling survey: optimization, Markov Decision Process (MDP), game-theory, Lyapunov, ML	Latency/energy models by class; unified notation within categories		Methodology-centric discussion	SDN context where relevant
Pashazadeh et al. (2023) [17]	MEC-vehicular interplay; placement, mobility, slicing	Frequent latency formulations; remarks on energy reporting		Sustainability notes within system design	SDN focus; Digital Twin (DT) emerging in scenarios
Ahmed et al. (2022) [32]	Vehicular task offloading; classification, issues, challenges	Task latency/throughput indicators; energy coverage by study		Performance-centric narrative	Selected SDN use-cases in vehicular settings
<i>This survey (2025)</i>	Mechanism-first, cross-layer taxonomy for C-/NR-V2X. DRL/MADRL focus. Evidence-weighted comparisons.	Unified latency-energy model across radio/compute/cache. Clear system vs vehicle bounds.		Aligned to SDG 7/11/13. Energy → CO ₂ per task. Congestion → air-quality pathways. Fairness indicators.	Integrates blockchain, digital twins, RIS-V2X, SDN, UAV relays, edge caching, carbon-aware scheduling.

This survey advances the field by (i) framing offloading decisions within sustainability and SDG-relevant indicators, (ii) proposing a simple yet unifying KPI model that normalizes latency/energy across heterogeneous resources, (iii) systematically reviewing emergent enablers (blockchain, digital twins, RIS, SDN, carbon-aware scheduling) under a method-matched lens, and (iv) articulating verification practices that separate targets from demonstrated achievements. This survey presents the following main contributions:

- **A sustainability-oriented taxonomy of AI-driven data offloading strategies:** We present a comprehensive and up-to-date classification of AI-enhanced data offloading approaches within vehicular networks, divided into infrastructure-based, vehicle-assisted, and hybrid models. Our taxonomy integrates perspectives from communication architecture, learning paradigms,

and system granularity, while explicitly linking each category to potential contributions to SDG 11 (Sustainable Cities and Communities) and SDG 13 (Climate Action). This provides a clear roadmap for researchers and practitioners aiming to design offloading systems that improve network efficiency while reducing energy waste and emissions.

- **In-depth comparative analysis of AI paradigms with sustainability metrics:** We analyze and compare state-of-the-art AI approaches, including supervised learning, unsupervised learning, reinforcement learning, and hybrid models, against critical performance indicators such as latency, throughput, energy efficiency, offload success ratio, and QoS satisfaction. Special emphasis is placed on Deep Reinforcement Learning (DRL), federated learning, and multi-agent systems in vehicular edge

environments, assessing not only technical performance but also their impact on environmental and operational sustainability.

- **Integration pathways with emerging technologies for resilient and green mobility:** We examine the convergence of AI-enabled offloading with 5G/6G networks, Software-Defined Networking (SDN), Network Function Virtualization (NFV), Mobile Edge Computing (MEC), Blockchain-based security, and digital twin frameworks. The discussion highlights how these technologies can synergize with AI to improve scalability, resilience, privacy, and energy efficiency, key enablers of sustainable and low-carbon urban vehicular communication systems.
- **Identification of open challenges and sustainability-aware research directions:** We systematically discuss unresolved issues that hinder large-scale deployment of intelligent offloading systems, including the explainability and interpretability of AI models, data heterogeneity, real-time constraint handling, energy–accuracy trade-offs, privacy risks in collaborative AI, and interoperability across multi-vendor vehicular systems. We also outline research priorities that incorporate sustainability indicators and life-cycle impact assessments to ensure that AI-driven offloading solutions contribute meaningfully to long-term smart city goals.

B. PAPER ORGANISATION

The remainder of the paper is organised as follows. Section II reviews the fundamentals of vehicular networks, data-offloading mechanisms, and the supporting smart-city communication infrastructure. Section III develops the proposed taxonomy of AI-driven offloading strategies, detailing infrastructure-based, vehicle-assisted, and hybrid solutions. Section IV conducts a comparative study of AI techniques for data offloading. Section V presents a comparative analysis of AI-Based Offloading Strategies with SDG Alignment. Section VI explores the integration of AI-powered offloading with emerging technologies and security frameworks. Section VII synthesizes the open challenges and research gaps and provides promising directions for future research. Finally, Section VIII concludes the paper.

II. FUNDAMENTALS OF URBAN VEHICULAR NETWORKS

Urban vehicular networks form the communication backbone of modern intelligent transportation systems (ITS). By enabling high-bandwidth, low-latency exchanges among vehicles, roadside infrastructure, and edge/cloud resources, they support a wide service spectrum: from safety-critical alerts and cooperative maneuvering to infotainment delivery and large-scale traffic optimization. Delivering these services in dense city centers demands stringent guarantees of latency, reliability, and seamless mobility, despite rapidly changing topologies and severe spectrum contention [28], [36], [37], [38].

A. ARCHITECTURE OVERVIEW (V2V, V2I, V2X, RSUS, OBUS)

Urban vehicular networks are typically organized into three interconnected layers that work together to enable vehicle-to-vehicle (V2V), vehicle-to-infrastructure (V2I) and overall vehicle-to-everything (V2X) communications. The foundational layer comprises On-Board Units (OBUs) installed in vehicles, which facilitate direct inter-vehicle connectivity, relay telemetry, transmit safety beacons, and provide information streams. These devices play a crucial role in the transmission of real-time information within high-traffic environments and represent the initial phase of the data plane [38], [39]. OBUs enable the swift and reliable distribution of information, enhancing traffic flow and reducing congestion, thus furthering the goals of SDG 11 (Sustainable Cities and Communities).

In the second tier, Roadside Units (RSUs) are placed methodically along roadways or used on mobile platforms, including buses or unmanned aerial vehicles. RSUs offer localized edge computing functionalities, manage medium-range vehicular clusters, and serve as intermediaries between OBUs and the core network [40], [41]. Their function in minimizing communication delays and facilitating energy-efficient routing leads to a decrease in petroleum usage and emissions, aligning with SDG 13 (Climate Action).

High-capacity edge and cloud platforms integrate vehicular data, execute global optimization and orchestration services, and support compute-intensive applications within the tertiary layer. Incorporation of mobile edge computing (MEC) nodes, often located in proximity to RSUs, minimizes the need for large-scale data transfers, leading to notable improvements in network energy efficiency and a decrease in latency [36]. Concurrent processing at a localized level reduces the environmental footprint associated with data-heavy vehicle applications while enhancing the quality of service provided.

Recent studies further advocate the organization of RSUs in self-organizing mesh overlays and the deployment of mobile relays to improve link robustness and maintain service continuity in obstructed urban corridors or during temporary network partitions [40], [41]. These architectural adaptations improve resilience to infrastructure failures, reduce the likelihood of congestion-induced emissions, and strengthen the sustainability of urban mobility ecosystems.

B. COMMUNICATION TECHNOLOGIES: DSRC, C-V2X, 5G/6G

Contemporary urban vehicular networks are built on a succession of radio-access families, each pushing stricter service guarantees. DSRC (IEEE 802.11p) supports single-hop V2V/V2I communication with minimal infrastructure and, under light traffic, can deliver one-way latency below 10 ms. In dense scenarios, carrier-sense multiple access with collision avoidance (carrier-sense multiple access with collision avoidance (CSMA/CA)) contention, hidden terminals, and limited spatial reuse substantially worsen delay

and reliability [39]. C-V2X improves LTE/5G by using sidelink and uplink modes, resulting in improved range and reliability. Field trials demonstrate a notable increase in packet delivery ratios compared to DSRC in non-line-of-sight urban environments [42].

The combination of 5G NR-V2X and C-V2X facilitates the incorporation of ultra-reliable low-latency communications (URLLC), network slicing, and improved MEC integration. Prototypes have shown notable performance, attaining one-way delays in the millisecond range and multi-gigabit throughput, which are critical for applications such as cooperative perception and platooning [43].

In the future, research into beyond-5G and 6G focuses on mmWave and THz operation, intelligent reflecting surfaces, and satellite or UAV-assisted backbones to achieve continuously connected and ultra-dense vehicular ecosystems [42]. A consolidated comparison of the latency, range, scalability, and maturity of the deployment of these technologies is provided in Table 2.

C. URBAN TRAFFIC DYNAMICS AND DATA GENERATION PATTERNS

Dense urban traffic exhibits highly variable mobility patterns stop-and-go waves, heterogeneous fleets (cars, buses, micro-mobility), and intermittent bottlenecks at complex intersections. These dynamics trigger bursty Basic Safety Message (BSM) traffic that requires end-to-end latencies below 100 ms with high reliability to prevent collisions [44]. In contrast, free-flow intervals produce an asymmetric, time-varying load across road segments and RSUs.

Modern connected and autonomous vehicles (CAVs) produce more than 25 GB/h⁻¹ raw sensor data (e.g., LiDAR, cameras, radar, ultrasound), many of which must be offloaded to edge servers or proximate peers for collaborative perception [28]. Priority scheduling is therefore essential to separate delay-critical streams (e.g., forward-collision warnings) from best-effort traffic (e.g., infotainment).

D. KEY TRANSMISSION CHALLENGES IN URBAN SETTINGS

There are five interconnected transmission challenges facing urban vehicular networks. Initially, the volatility of the links and the dynamic nature of the topology, which are characterized by rapid acceleration, braking, and lane changes, substantially disrupt the wireless continuity. Consequently, real-time mobility prediction is essential for effective routing, as it reduces route interruptions and minimizes handover delays [38], [45]. Secondly, the presence of severe multipath fading and non-line-of-sight conditions is significant; urban canyons generate intricate multipath environments and encounter frequent obstructions. Implementing countermeasures such as spatial diversity MIMO, adaptive beamforming, and multi-antenna RSUs improves both reliability and throughput [44]. Third, spectrum congestion and coexistence pose substantial obstacles. The 5.9 GHz ITS band is in use

by DSRC, C-V2X, Wi-Fi, and emerging NR-U links, which could result in a 40% decrease in packet delivery ratios in densely populated environments lacking coordination. Although cognitive sensing and dynamic channel selection present potential solutions, the regulatory harmonization process is still in the early stages of development [39]. Fourth, the issue of QoS heterogeneity arises: ultralow latency safety traffic must coexist with bandwidth-intensive services, such as HD maps and video. Edge-intelligent RSUs that facilitate real-time flow classification, priority queueing, and proactive caching are instrumental in addressing these conflicting requirements [40], [42]. Fifth, the economic implications of the deployment of RSU reveal that dense RSU grids can lead to reduced latency; however, they also result in increased capital and operational expenditures. In contrast, hybrid topologies that incorporate a combination of fixed RSUs and mobile relays, such as buses or UAVs, have the potential to improve effective coverage by 30% and decrease end-to-end latency by 20% compared to static deployments [41].

In order to effectively address these challenges, it is imperative to adopt a comprehensive approach. This involves the integration of AI-assisted control techniques, such as proactive handovers and data offloading, with advancements at the physical layer, such as adaptive modulation and beamforming. In order to meet the performance objectives established for the next iteration of Intelligent Urban Transportation Systems, it is necessary to collaborate on strategic infrastructure planning, which encompasses the optimization of RSU placement and relay scheduling.

E. TASK ENERGY MODEL AND EMISSIONS ACCOUNTING

We quantify the energy per vehicle task with an explicit separation between baseline (idle/housekeeping) and incremental components (transmit/receive/compute/state), aligned with the mobile/edge energy accounting in the ICT literatures [22], [46], [47], [48], [49], and [50].

$$E_{\text{tot}} = P_{\text{idle}} t_{\text{task}} + E_{\text{tx}} + E_{\text{rx}} + E_{\text{proc}} + E_{\text{state}}, \quad (1)$$

Transmission and reception energies are

$$E_{\text{tx}} = (P_{\text{tx}}(\gamma_{\text{ul}}) + P_{\text{oh,tx}}) t_{\text{tx}}, \quad E_{\text{rx}} = (P_{\text{rx}} + P_{\text{oh,rx}}) t_{\text{rx}}, \quad (2)$$

with durations linked to observable rates

$$t_{\text{tx}} = \frac{D_{\text{ul}}}{r_{\text{ul}}}, \quad t_{\text{rx}} = \frac{D_{\text{dl}}}{r_{\text{dl}}}, \quad r_{\bullet} = B_{\bullet} \log_2(1 + \gamma_{\bullet}). \quad (3)$$

For on-board processing with L CPU cycles at frequency f (DVFS model):

$$t_{\text{proc}} = \frac{L}{f}, \quad E_{\text{proc}} = \kappa L f^2 + P_{\text{oh,proc}} t_{\text{proc}}. \quad (4)$$

If sleep/idle/deep-sleep states are used, transitions incur

$$E_{\text{state}} = \sum_{(p \rightarrow q) \in \mathcal{S}} n_{pq} E_{pq}. \quad (5)$$

TABLE 2. Comparison of DSRC, C-V2X, and 5G/6G V2X communication technologies.

Feature	DSRC 802.11p)	(IEEE C-V2X (3GPP Rel. 14/15)	5G/6G V2X (3GPP Rel. 16+)
Standardization Body	IEEE	3GPP	3GPP
Frequency Band	5.9 GHz ITS band	5.9 GHz ITS band (PC5); licensed LTE bands (Uu)	FR1 (sub-7.125 GHz), FR2 (24.25–52.6 GHz); exploratory sub-THz (6G)
Typical Latency	10–50 ms (E2E, typical)	10–20 ms (PC5 sidelink, typical)	< 1ms (URLLC air interface target); ~ 1–10 ms E2E (use-case dependent); 6G aims sub-ms
Communication Modes	V2V, V2I	V2V (PC5), V2I, V2N (Uu), V2P	V2V, V2I, V2N, V2P (enhanced NR-V2X)
Infrastructure Dependency	Low (ad hoc)	Low–Moderate (ad hoc PC5; optional network assist)	Moderate (NR sidelink possible; many features edge/core-enabled)
Maximum Range	300–1000 m (LOS)	Up to ~1–2 km (LOS)	Sidelink: hundreds of m to ~1–2 km (LOS); FR1 cellular coverage: several km
Scalability	Limited (MAC contention)	Improved (sidelink scheduling/SPS)	High (network slicing, NR sidelink modes, edge orchestration; 6G: RIS)
Deployment Maturity	Limited/commercial pilots	Commercial rollouts underway	Trials and pilot deployments
Key Features	Low-latency ad hoc without infrastructure	Cellular integration, better reliability/coverage	URLLC, MEC, slicing, massive MIMO; towards AI-native 6G

Partial offloading to edge servers is represented by $\theta \in [0, 1]$:

$$L_{\text{loc}} = (1 - \theta)L, \quad D_{\text{ul}}(\theta) = \theta D_{\text{ul}}^*, \quad D_{\text{dl}}(\theta) = \theta D_{\text{dl}}^*. \quad (6)$$

When the system boundary includes radio/access/backhaul and edge, we report

$$E_{\text{system}}(\theta) = E_{\text{veh}}(\theta) + \alpha E_{\text{RAN+BH}}(\theta) + \beta E_{\text{edge}}(\theta), \quad (7)$$

with allocation factors α, β consistent with telecom energy-efficiency guidance [49], [50].

Sustainability reporting uses energy per delivered bit and task-level emissions:

$$e_{\text{bit}} = \frac{E_{\text{tot}}}{D_{\text{ul}} + D_{\text{dl}}}, \quad C_{\text{task}} = \lambda_{\text{grid}} \frac{E_{\text{system}}}{3.6 \times 10^6}, \quad (8)$$

where λ_{grid} is the contemporaneous emission factor of the grid (kgCO₂e/kWh); the denominator converts joules to kWh. We document measurement boundaries (vehicle-only vs. system), parameter sources, and baselines to enable replication.

F. INTEROPERABILITY BETWEEN CONVENTIONAL, HYBRID, AND ELECTRIC VEHICLES

From the perspective of vehicular edge computing and AI-based offloading, current urban deployments already operate over highly heterogeneous fleets that mix conventional internal-combustion vehicles, hybrid vehicles, and battery electric vehicles (EVs). At the communication layer, most vehicles share common V2X stacks (e.g., C-ITS/DSRC or C-V2X/NR-V2X) and in-vehicle interfaces (CAN, OBD-II), so data exchange and access to basic state information (speed, position, sensor readings) are largely interoperable across powertrain types. The main differences arise in application-layer services: EVs expose additional state variables (state-of-charge, predicted range, charging schedule) and interact with dedicated charging infrastructure, while conventional and hybrid vehicles primarily use fuel-related diagnostics. In practice, VEC and AI-based offloading frameworks can treat all vehicle classes as a common pool of mobile nodes for safety and traffic-efficiency services, while exploiting EV-specific information for energy-sensitive routing, smart charging, and demand-response applications. This heterogeneity motivates the need for offloading policies and data

models that remain agnostic to powertrain technology at the networking layer, but can incorporate EV-specific capabilities when optimizing sustainability-oriented services [51].

III. DATA OFFLOADING IN VEHICULAR NETWORKS: CONCEPTS AND TAXONOMY

In dense urban centers, connected and autonomous vehicles generate continuous data streams from heterogeneous sources, including LiDAR, radar, camera arrays, vehicular telematics, and user applications. During rush hour platoons or in the aftermath of traffic incidents, the resulting surge in traffic can overwhelm cellular links, increase latency, cause packet loss, and increase energy consumption [52], [53]. Inefficiencies reduce service quality while simultaneously increasing carbon emissions and congestion, undermining the goals of SDG11 (Sustainable Cities and Communities) and SDG13 (Climate Action).

To address these bottlenecks, data offload reallocates traffic via alternative communication pathways. Roadside Units (RSUs) serve as local access points and MEC gateways, effectively managing traffic surges and executing computations close to the source. This approach minimizes the need for long-distance transmissions and contributes to reduced energy consumption. V2V and D2D are examples of peer-to-peer links that facilitate the direct exchange of safety beacons, sensor maps, and other critical information without requiring a transit through the core network. This reduces transmission overhead and latency. Mobile Edge Computing (MEC), which is located in close proximity to RSUs or base stations, effectively manages latency-sensitive tasks such as object detection and path planning within a single round-trip time of the vehicle. This accelerates decision-making and reduces power consumption.

When effectively coordinated, these diverse routes minimize end-to-end delay, optimize battery usage, and ensure service continuity in regions with sporadic coverage or inconsistent infrastructure, while also mitigating emissions associated with congestion. Figure 2 illustrates a cohesive architecture that incorporates RSUs, V2V relays, and cloud-edge servers.

Recent research has shown that AI-guided policies that implement mobility prediction, link quality estimation, and task criticality consistently outperform static heuristics in terms of throughput and delay [54]. For example, DRL agents have achieved double digit reductions in queueing delays under bursty workloads by jointly optimizing RSU association and V2V relaying decisions. This has resulted in lower fuel consumption as a result of reduced idling periods.

We introduce a taxonomy that classifies vehicular data offloading schemes along three principal dimensions to provide a structured view of the literature. The initial dimension outlines infrastructure dependency, distinguishing between infrastructure-centric models that rely on fixed components like RSUs or MEC nodes, vehicle-assisted models that function solely through V2V or D2D connections, and hybrid strategies that flexibly combine both elements.

The second dimension pertains to the level of decision-making intelligence, which encompasses rule-based heuristics, optimization-driven approaches, and AI-powered strategies, such as supervised learning, reinforcement learning (RL), and multi-agent systems. The third dimension differentiates between single-hop offloading, multi-hop relaying, and opportunistic or disruption-tolerant routing mechanisms, taking into account the underlying communication model. In terms of energy efficiency, resilience to urban mobility and reduction of the carbon footprint, these classifications establish a basis for evaluating the sustainability impact of each approach.

A. DEFINITIONS AND KEY METRICS

In particular, in smart urban environments that are dynamic, bandwidth-constrained, and latency sensitive, key performance indicators (KPIs) are indispensable to assess the efficacy of vehicular data offloading strategies. The potential impact of these KPIs on energy consumption, resource utilization, and carbon emission factors directly related to **SDG 11** (Sustainable Cities and Communities) and **SDG 13** (Climate Action) is not the only consideration in the design of sustainable systems; they are also related to network efficiency. Table 3 presents the main ITS-aligned KPIs and their relevance to vehicular data offloading.

Figure 3 summarizes the overall pipeline of AI-based vehicular offloading. On the left, heterogeneous vehicular data (e.g., positions, speeds, sensed environment, application requirements) are combined with real-time network state information (channel quality, RSU load, coverage, and interference) to form the context observed by the controller. In the central block, AI-based offloading modules—ranging from classical ML to deep RL and federated/distributed learning jointly decide *where* each task should be executed (on-board, at the RSU/MEC, or in the cloud) and *how* radio, compute, and caching resources are allocated across flows. On the right, these decisions translate into the key performance indicators of the vehicular network, including end-to-end latency, energy efficiency, reliability, and throughput, as well as their aggregate impact on smart-city sustainability targets and SDG-oriented outcomes.

The primary KPIs considered in this survey are as follows:

End-to-End Latency: The total round-trip delay between the creation of data at the OBU and the completion of the task at the processing node. Time-critical functions, such as forward collision warning, require a latency of less than 10 milliseconds. By proactive redirecting traffic to lightly burdened RSUs and predicting imminent link failures, AI-enabled policies improve this metric [1], [2], [55].

Delay Jitter and Deadline Miss Ratio: The variance in delay and the probability of exceeding packet deadlines are equally important; mean latency alone does not guarantee real-time performance. In recent research, probabilistic delay bounds that are resilient to bursty traffic and rapid topology variation have been derived [46], [56].

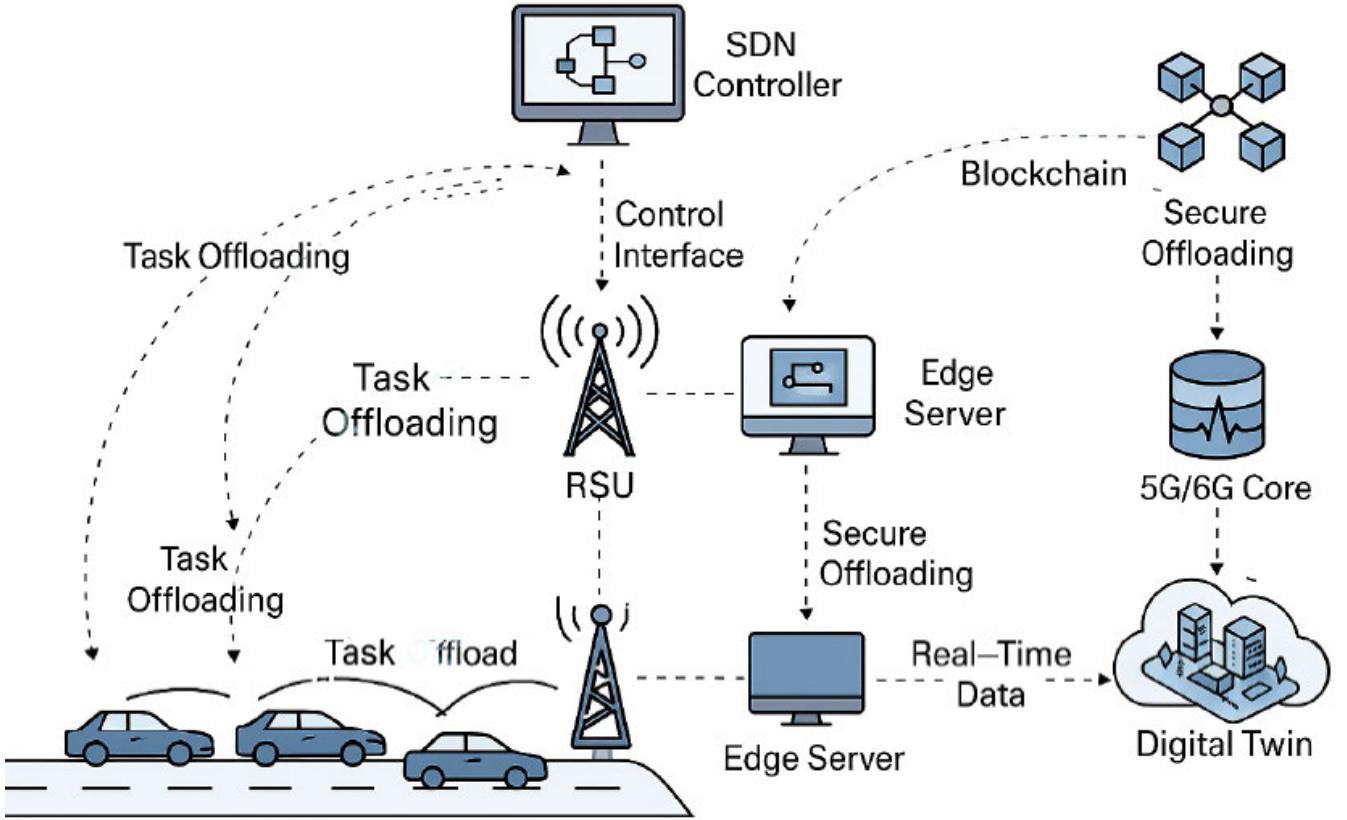


FIGURE 2. Integrated architecture for AI-based offloading in smart vehicular networks.

Bandwidth (Spectral) Efficiency: The ratio of delivered payload bits to total transmitted bits, including headers and retransmissions. Cross-layer offloading schemes dynamically allocate DSRC, C-V2X, and mmWave resources to maximize spectral efficiency while minimizing control overhead [57].

Energy Consumption: The total radio and computational energy expended per completed task or per Mbit delivered. Compared to fixed-threshold baselines, DRL-based controllers that jointly adapt the transmit power, modulation, and payload path have reported energy savings of up to 2–5% [22], [46], thus contributing to reduced carbon emissions in the process.

Offloading Ratio: The percentage of data or tasks that were effectively diverted from the cellular macro-link to auxiliary paths, such as RSU, V2V, or D2D. The effective use of peripheral infrastructure, reduced congestion in the core network, and potential energy savings are indicators of a high transfer ratio [9].

Quality of Service Satisfaction Rate: The fraction of traffic flows that meet predefined service thresholds, for example, latency ≤ 10 ms, reliability $\geq 99.9\%$ and throughput ≥ 10 Mbps. AI-based schedulers improve this rate by prioritizing safety-critical flows while deferring best-effort traffic [3].

Quantitative models are used to jointly analyze latency and energy consumption. The total latency is expressed as:

$$D_{\text{tot}} = D_{\uparrow} + D_{\text{edge}} + D_{\downarrow}, \quad (9)$$

$$D_{\uparrow} = \frac{S}{R_{\uparrow}}, \quad D_{\text{edge}} = \frac{C}{f_e}, \quad D_{\downarrow} = \frac{S_{\downarrow}}{R_{\downarrow}}, \quad (10)$$

where S is the size of the offloaded data (in bits), $R_{\uparrow}$ is the uplink data rate (bps), C is the total number of CPU cycles required for task execution and f_e denotes the operating frequency of the edge server CPU (Hz). This latency decomposition follows the bimodal model (communication plus computation delay) formalized in [1] and [2].

The total energy consumption per vehicular task, which includes transmission, reception, and edge processing, is given below.

$$E_{\text{tot}} = \sum_{i \in \{\text{tx}, \text{rx}, \text{proc}\}} P_i t_i, \quad (11)$$

where P_i denotes the instantaneous power (W) and t_i the operating time (s) of phase i . This formulation aligns with the energy accounting frameworks of the different layers reported in [22] and [46] and can be extended to incorporate sustainability metrics such as the carbon cost per bit delivered.

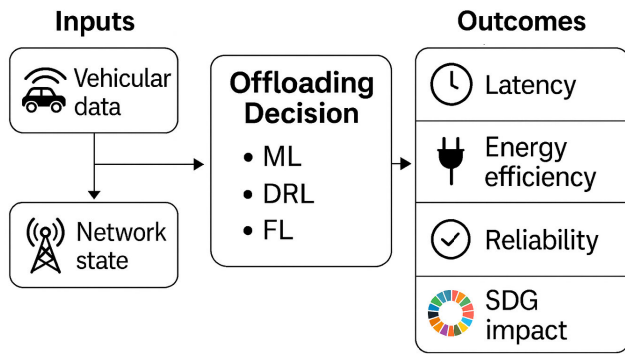


FIGURE 3. High-level conceptual view of AI-based vehicular offloading. Vehicular data and network state are fed into AI-based offloading decision modules (e.g., ML, DRL, FL), which generate actions that impact key KPIs (latency, energy efficiency, reliability) and ultimately SDG-oriented sustainability outcomes.

B. TAXONOMY OF DATA-OFFLOADING APPROACHES

Vehicular data offloading schemes can be categorized into three overarching paradigms, differentiated by their reliance on fixed infrastructure and by the locus of decision-making intelligence. The main paradigms, along with their representative subcategories, are illustrated in Figure 4.

1) INFRASTRUCTURE-BASED APPROACHES

In infrastructure-based schemes, fixed assets, including cellular base stations, roadside units (RSUs), and co-located mobile edge computing (MEC) servers, are responsible for data uploading and task execution. Sensor frames or computational tasks are typically transmitted by vehicles to the nearest RSU, where they are processed and global resource orchestration is performed at the edge or cloud tier. Centralized control improves the ability to implement precise admission control and load balancing, thus improving the adherence to quality of service standards in congested urban environments. However, performance is compromised in areas with insufficient RSU coverage or due to infrastructure malfunctions [57], [58], [59]. To address these vulnerabilities, recent research has integrated blockchain-based accountability frameworks and AI-driven scheduling modules into RSU clusters, thus improving impartiality and decreasing the number of single failure points. Designs contribute to network resilience, aligning with the sustainability objectives of SDG 11 by enabling the delivery of reliable urban mobility services.

2) VEHICLE-ASSISTED APPROACHES

Vehicle-assisted methods employ V2V and D2D links to enable data exchange or task execution for adjacent vehicles, thus transforming specific vehicles into mobile ad hoc RSUs. This collaboration guarantees service continuity in regions with inadequate infrastructure or those affected by disasters, while simultaneously improving the efficiency of spatial spectrum utilization [28], [60]. However, the framework must confront the trust and security challenges that result from

interactions with unverified peers, as well as the issues associated with high mobility and intermittent connectivity. When implemented effectively, these strategies reduce dependence on existing infrastructure, reduce communication-related energy expenditures, and improve coverage. Consequently, they contribute to SDG 11 by improving accessibility and to SDG 13 by reducing emissions through optimized routing.

3) HYBRID APPROACHES

Hybrid schemes integrate infrastructure-centric and vehicle-assisted strategies, adapting offloading decisions to task criticality, link quality, network congestion, and real-time RSU availability. Deep reinforcement learning (DRL) agents are frequently used to optimize these decisions, resulting in statistically significant reductions in end-to-end latency and energy consumption relative to static baselines [15], [21], [61]. This adaptability enables more efficient task distribution and improves service quality in diverse network conditions.

A notable subclass of hybrid solutions employs unmanned aerial vehicles (UAVs) to extend coverage in infrastructure-limited or disaster-affected regions. UAVs, equipped with MEC capabilities, can vary in energy budgets, flight patterns, and onboard computational power. Yan et al. [62] propose RUDDER, a delayed feedback scheduling framework for UAV swarms, which reduces average data queue lengths by 35% and increases UAV operational time by 28%. Similarly, Apostolopoulos et al. [63] integrate MEC into UAVs and frame the offloading process as a Prospect-Theory-based game, enabling stable task distribution under energy constraints while lowering latency and improving user satisfaction. These context-aware orchestration strategies not only improve operational resilience, but also advance the energy efficiency and low-emission objectives of SDG 13, especially in challenging connectivity environments.

C. OFFLOADING SCENARIOS IN SMART-CITY USE-CASES

Smart-city services rely on vehicular networks to facilitate mobility, enhance situational awareness, coordinate public safety, and support large-scale environmental sensing; in this context, adaptive offloading is critical for preserving Quality of Service while alleviating cellular backhaul load. Vehicles operate as dynamic sensors that report speed, density, and congestion to roadside units (RSUs), where strategic RSU placement improves coverage [41], routing-aware protocols capture urban mobility patterns [44], and deep reinforcement learning (DRL)-assisted relaying balances traffic under varying conditions [64]. For infotainment and content delivery, mobile edge computing (MEC) caching combined with vehicle-to-vehicle (V2V) sharing sustains high-bit-rate streaming, with attention-based DRL path selection improving quality of experience (QoE) [65] and energy-sensitive controllers cutting power by 2–5% without degrading playback [22]; joint V2I/V2V prefetch strategies further benefit ride-sharing fleets [27], [66]. Handover and mobility management benefit from proactive prediction:

TABLE 3. ITS-aligned KPIs and their relevance to vehicular data offloading.

KPI	Definition / Unit	Typical Target	Why it matters for offloading
E2E Latency	Source-to-sink delay [ms]; report mean and tail (p99)	< 10–100 ms (use-case dependent)	Safety BSM deadlines, cooperative perception freshness; offloading adds uplink + compute + downlink terms.
PDR/Reliability	Packet Delivery Ratio [%] / outage [%]	$\geq$ 99.9% (safety)	Robustness under blockage/canyons; scheduling and path selection.
Throughput	Accepted app-layer rate [Mb/s]	Scenario-dependent	Map/vision flows; split between OBU/RSU/cloud with MEC support.
Energy per bit	J/bit at OBU+RSU+edge	Lower is better	Captures radio + compute cost of offloading vs. local execution.
Carbon per bit	$g\text{CO}_2\text{e}/\text{bit}$	Lower is better	Converts energy to carbon via grid intensity; informs sustainability.

DRL anticipates link-quality drops and triggers handovers to reliable RSUs or cooperative relays, reducing loss and latency spikes [26], while multi-agent coordination minimizes disruptions in dense deployments. Public-safety applications (e.g., emergency braking and collision avoidance) demand ultra-reliable, low-latency communication; hybrid infrastructure with V2V relaying helps maintain delivery under partial outages [2], multi-agent DRL improves relay selection [15], and coalition-based spectrum sharing prioritizes safety-critical traffic during congestion [67]. For environmental sensing and urban monitoring, vehicular crowdsourcing gathers air-quality, noise, and microclimate data; machine-learning schedulers limit unnecessary uploads while meeting freshness constraints [24], visible-light V2V links offer an unlicensed option for non-critical data [60], and fog/edge frameworks provide scalable, energy-efficient processing [15], [68], [69]. Finally, routing-integrated offloading jointly optimizes forwarder selection and load paths, leveraging road topology and RSU availability [70] and dynamically switching between V2V relays and RSU uplinks according to hop-wise delay estimates [71]. Representative AI-enabled offloading use cases in smart-city vehicular networks including traffic monitoring, infotainment, public safety, and cooperative driving are summarized in Table 4, which details strategies, performance benefits, and deployment considerations.

D. NETWORK PERFORMANCE: LATENCY AND OUTAGES

Because vehicular edge computing is deployed in highly dynamic wireless environments, a clear view of network performance is essential before analyzing AI-based offloading strategies. In this survey, we focus on two key dimensions: end-to-end latency and outages (service unavailability / reliability degradation).

1) END-TO-END LATENCY

We define end-to-end latency as the total delay experienced by an offloaded task from generation at the vehicle to

reception of the result. This includes (i) uplink transmission and access delay (vehicle $\rightarrow$ RSU/base station), (ii) backhaul and core-network delay (RSU/base station $\rightarrow$ edge or cloud server), (iii) queuing and processing delay at the MEC server, and (iv) downlink delivery of the results to the vehicle. For URLLC-like safety applications, the relevant KPI is often the tail of the latency distribution (e.g., 95th or 99th percentile latency), rather than the mean, since rare spikes can compromise safety and user acceptance.

2) OUTAGES AND RELIABILITY

Outages can be characterized at several levels: (i) link-level outages, when the instantaneous signal-to-interference-plus-noise ratio (SINR) falls below a threshold and packets are dropped; (ii) service outages, when the effective latency exceeds an application-specific deadline; and (iii) coverage gaps, when no suitable RSU/MEC node is available.

These effects are often summarized via reliability indicators, such as the probability that latency stays below a target deadline, the fraction of tasks successfully served, or the percentage of time that the VEC service is available within a region. In the context of smart cities and SDG-oriented evaluation, such reliability metrics are directly linked to perceived quality of service for traffic safety, traffic efficiency, and critical information services.

3) RELATIONSHIP TO THE KPI MODEL

The minimal KPI model introduced earlier decomposes overall system performance into radio, compute, and caching components. Latency and outage probabilities are derived by aggregating these components across the offloading path and by accounting for handover events, congestion, and failures. The quantitative results reported in later sections are interpreted with respect to this view of network performance, so that the improvements claimed by AI-based offloading strategies can be consistently compared across studies.

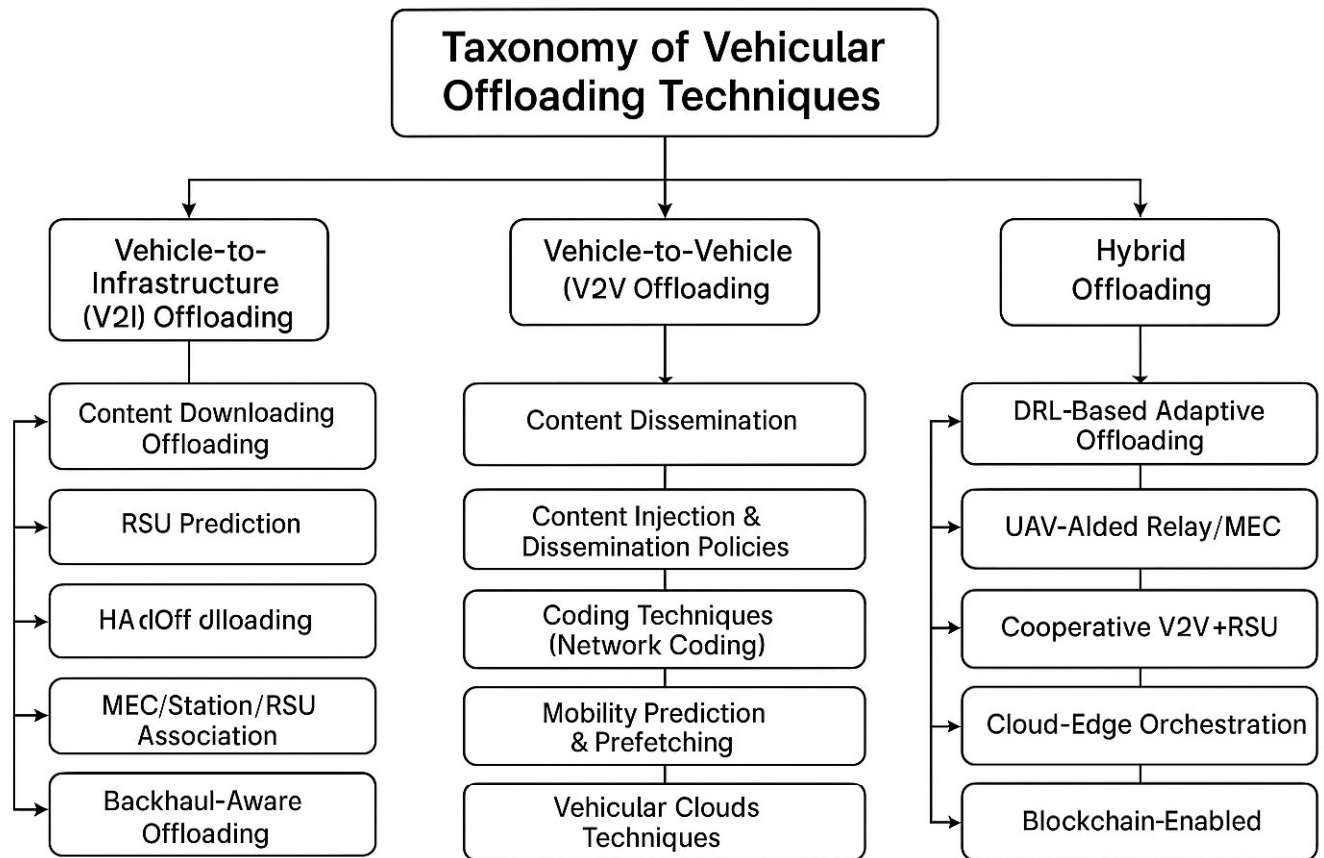


FIGURE 4. Taxonomy of vehicular data-offloading techniques.

IV. AI TECHNIQUES FOR DATA OFFLOADING

Artificial intelligence (AI) has emerged as the cornerstone of modern data-offloading frameworks for urban vehicular networks. The inherent heterogeneity of these systems, marked by volatile mobility patterns, rapidly varying channel conditions, and stringent application-specific QoS requirements, renders static or heuristic strategies increasingly inadequate. In contrast, AI models can process high-dimensional state information in (near) real-time, uncover latent correlations, and synthesize actions that are optimal or near-optimal under uncertainty. This section reviews the dominant AI paradigms: classical supervised and unsupervised learning, deep reinforcement learning (DRL), privacy-preserving federated learning (FL), and emerging meta-learning approaches highlighting their capabilities, limitations, computational overheads, and deployment considerations. Beyond describing each paradigm in isolation, this section also contrasts their relative strengths, weaknesses, and typical roles in vehicular edge-computing (VEC) architectures.

Previous studies mainly used supervised classifiers to predict link stability or classify offloading contexts into discrete decision spaces [29]. More recent methodologies employ DRL to learn sequential actions and capture long-term returns that static policies cannot model [54], [81]. FL enhances these capabilities by distributing model training across vehicles,

thus preserving data locality and ensuring the convergence of robust global models [80]. The “learning-to-learn” paradigm of meta-learning in vehicular environments, despite its early stage, offers the potential for rapid adaptation to novel traffic regimes and deployment contexts with minimal fine-tuning data.

A. MACHINE-LEARNING FOUNDATIONS (SVM, DECISION TREES, RANDOM FORESTS)

Supervised learning algorithms, including Support Vector Machines (SVM), Decision Trees (DT), and Random Forests (RF), are fundamental because to their minimal inference latency, limited memory requirements, and straightforward implementation on embedded systems.

These models correlate observable network-state characteristics (e.g., received signal strength, RSU load, vehicle density, queue length) with specific offloading options. Support Vector Machines (SVMs) have notable efficacy in high-dimensional environments with constrained training datasets and have been utilized in delay-sensitive scheduling and intrusion detection within Vehicular Ad Hoc Networks (VANETs), where sub-millisecond inference is essential [36]. Decision Trees provide clear, rule-based decision boundaries, making them appropriate for safety-critical applications that require interpretability, whereas Random Forest ensembles

TABLE 4. AI-enabled offloading use cases for smart-city vehicular networks.

Use Case	Offloading Strategy	Quantified Benefits & Key Considerations	References
Traffic Monitoring	V2V relay of congestion states; RSU-assisted MEC aggregation with load-aware routing; DRL-based dynamic RSU selection	Mean intersection delay reduced by 30–47% and throughput improved by 25% in SUMO/VEINS simulations; requires privacy-preserving geo-location sharing and secure V2V authentication.	[46], [66], [72], [73]
Infotainment	Edge caching of popular media at RSUs; multi-agent RL-guided V2V multimedia dissemination; trajectory-aware D2D prefetching	Playback stall time reduced by 21% and backhaul load decreased by 26% in real-trace experiments; dependent on accurate mobility prediction and content verification mechanisms.	[53], [74]–[76]
Public Safety (URLLC)	Prioritised BSM broadcast; RSU relay to emergency responders; event-triggered MADRL scheduling	End-to-end latency < 100 ms with 95% reliability using 24% fewer DRL updates; requires multi-agent trust and fairness enforcement.	[15], [61], [77]
Environmental Sensing	Opportunistic vehicle-borne sensing with periodic RSU upload; V2V aggregation in sparse-RSU areas	City-wide crowd-sensing achieved with 35% fewer static sensors compared to fixed deployment; sensor calibration remains a primary challenge.	[62], [63]
Smart Parking	RSU broadcast of space availability; V2V exchange of nearest free slots; DRL-based reservation and dynamic pricing	Search time reduced by 25–30% and CO ₂ emissions decreased by 12% compared to heuristic approaches; privacy maintained through federated learning.	[78]–[80]
Co-operative Driving	Low-latency V2V platoon coordination; RSU-assisted route re-planning under congestion; UAV-augmented offloading with mobility prediction	Fuel consumption reduced by 13–18% and headway variance reduced by 35%; requires reliable mobility-aware handovers.	[56], [56]

reduce the variance of individual trees and improve generalization (for instance, selecting between RSU and cellular paths in uncertain channel conditions) [68].

Despite their advantages, classical models struggle with the non-stationarity and long-range temporal dependencies characteristic of urban mobility. As a result, they are often used as baselines, feature preprocessors, or interpretable components within hybrid pipelines rather than as standalone solutions [28].

Recent extensions combine statistical learning with adaptive rule sets to refine context models and stabilize latency under topological changes [74]. In general, these learners

are most effective when interpretability, tight computational budgets, or deterministic execution times are primary concerns; deep learning or DRL should be considered when environmental dynamics invalidate fixed decision boundaries.

B. DEEP-LEARNING APPROACHES (CNNs, RNNs/LSTMS, GNNs)

Deep learning models capture multiscale spatiotemporal patterns in vehicular mobility and wireless dynamics. Convolutional neural networks (CNNs) distill local fluctuations

in path loss, interference, and blockage to enable rapid link quality estimation and microcell / handover selection.

In parallel, recurrent neural networks (RNNs) including variants of long-short-term memory (LSTM) model temporal dependencies to predict RSU association windows and the persistence of V2V links at high speeds [82], [83]. More recently, graph neural networks (GNNs) operate on dynamic vehicular graphs, where nodes denote OBUs/RSUs and edges denote communication links; by jointly exploiting node attributes (e.g., queue length, residual energy) and topology, these models optimize end-to-end offload paths and edge-server selection [75], [84].

Empirical findings indicate that deep models achieve superior prediction accuracy, faster adaption, and enhanced QoS assurances relative to CNN/LSTM baselines. However, their training requires extensive labeled data sets, considerable computing resources, and substantial memory, which may restrict direct implementation on OBUs [1], [85]. Practical mitigations encompass the construction of lightweight architectures or the application of knowledge distillation to adhere to hardware limitations, alongside the utilization of ongoing learning or domain adaptation to counteract performance degradation resulting from domain shifts (e.g., sim-to-real or inter-city transfer). In comparative terms, deep models typically provide richer state representations and higher accuracy than classical ML at the expense of interpretability and energy cost, and they often serve as perception or prediction back-ends whose outputs are subsequently exploited by DRL, FL, or meta-learning components.

C. REINFORCEMENT LEARNING AND DEEP Q-NETWORKS

Reinforcement Learning (RL) aims to optimize long-term rewards that are associated with latency, energy consumption, throughput, or task-success ratio by representing offloading as a sequential decision-making problem under uncertainty. Value-based methods, such as Deep Q-Networks (DQN), and actor-critic approaches such as Proximal Policy Optimization (PPO), A3C, and DDPG, are frequently implemented [2], [8]. An LSTM-augmented PPO agent, for example, can proactively prevent congestion by predicting the load of the base station, resulting in latency reductions exceeding 30%. Multi-Agent Deep Reinforcement Learning (MADRL) facilitates adaptation to spatial traffic gradients and diverse Mobile Edge Computing (MEC) capacities at scale by fostering decentralized yet coordinated policies [10], [15].

Although DRL demonstrates robust asymptotic performance, it also poses obstacles such as exploration overhead, intricate reward design, and stability issues in the context of non-stationarity or partial observability. Sample inefficiency persists as a constraint, prompting the adoption of curriculum learning, model-based rollouts, or offline reinforcement learning to reduce perilous online exploration. In safety-critical environments like URLLC, limited reinforcement learning or protected regulations are vital to avert hazardous exploratory actions. Compared with purely supervised

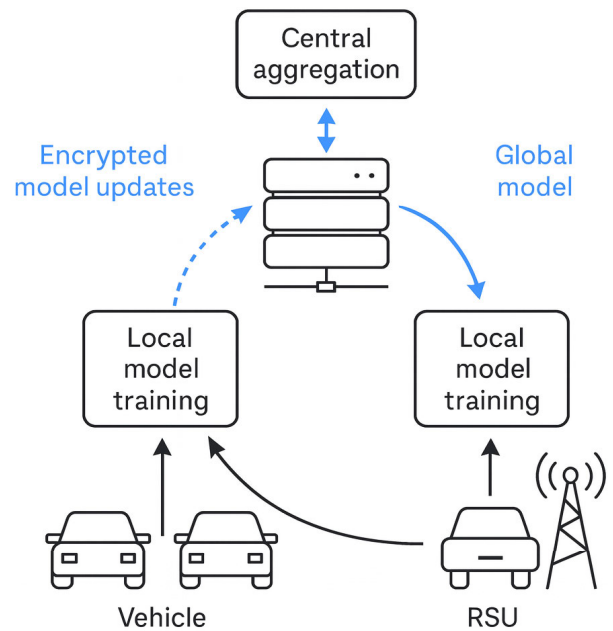


FIGURE 5. Privacy-preserving federated-learning workflow for vehicular offloading.

schemes, DRL can typically achieve larger long-term gains in latency and energy, but at the cost of more complex training pipelines, dependence on high-fidelity simulators, and more challenging safety certification.

D. FEDERATED AND DISTRIBUTED LEARNING FOR PRIVACY PRESERVATION

Federated Learning (FL) enables collaborative model training across vehicles and RSUs while retaining raw data locally, satisfying privacy and data-sovereignty requirements [77]. In standard FL, clients perform local updates that are then aggregated centrally; split-FL partitions computation between client and edge to reduce OBU load and uplink bandwidth usage, while secure aggregation prevents the reconstruction of individual contributions [86]. Advanced variants such as federated GNNs (FedGNN) tailor FL to dynamic vehicular graphs, optimizing age-of-information and offloading latency, while genetic-assisted FL re-weights client updates to counteract stragglers and non-IID heterogeneity under strict privacy and delay constraints [87]. The general workflow is depicted in Figure 5.

Although FL reduces the need for data sharing, it introduces new challenges related to communication and aggregation. Inconsistent or stale updates, client churn, and hardware heterogeneity can hinder convergence or bias results. Techniques such as hierarchical aggregation, client selection, and gradient compression help reduce communication overhead, while robust aggregation and differential privacy mitigate poisoning and inference attacks at the expense of some accuracy.

It is important to budget for the additional control plane traffic that FL introduces, particularly in dense urban environments with intermittent connectivity. Relative to centralized deep learning or DRL, FL therefore trades some convergence speed and model optimality for improved privacy, regulatory compliance, and scalability, and is often most effective when combined with edge/cloud training and inference pipelines.

E. META-LEARNING AND TRANSFER LEARNING

Meta-learning and transfer learning aim to accelerate adaptation when exhaustive retraining is impractical. Meta-learning develops an initialization or optimizer that generalizes across tasks, enabling rapid fine-tuning with minimal data, while transfer learning adapts models trained in one context (e.g., city or road network) for deployment in another [21]. Genetically optimized federated meta-learning can further reduce communication rounds while simultaneously minimizing latency and privacy leakage in 5G-enabled vehicular networks.

The benefits of rapid adaptation are balanced by the risks of negative transfer when source and target distributions diverge (e.g., differences in RSU density or mobility patterns). Selecting appropriate source tasks, applying regularization, and validating target data are necessary to preserve QoS. Combining meta-learning with FL offers a promising path toward city-to-city portability without compromising data privacy. In comparison with stand-alone DL or DRL approaches, meta- and transfer-learning are less focused on achieving the absolute best performance on a single scenario and more on preserving acceptable performance across multiple deployment contexts with limited additional data and computation.

In summary, supervised learners offer interpretability and predictable execution for embedded control, deep models extract complex spatio-temporal structure, DRL optimizes long-term KPIs, FL ensures data locality at scale, and meta-learning accelerates adaptation across domains. Effective systems often integrate these paradigms: lightweight predictors for fast screening, deep perception modules for rich state estimation, DRL for orchestrating resources under constraints, and federated or meta-learning to sustain performance under evolving urban conditions. A side-by-side comparison of these paradigms is provided in Table 5. From a comparative perspective, classical ML remains preferable when computational budgets, interpretability, and deterministic execution times are the primary constraints, while deep learning is advantageous when rich sensor data and complex spatio-temporal correlations dominate. DRL becomes most attractive when long-horizon trade-offs between latency, energy, and reliability must be optimized, but requires careful engineering of simulators and safety mechanisms. FL and distributed learning are particularly suited to jurisdictions with strong privacy requirements and fragmented data ownership, whereas meta-/transfer-learning are promising for operators that must deploy comparable offloading strategies across multiple cities or corridors

with heterogeneous infrastructure. In practice, state-of-the-art vehicular offloading solutions increasingly combine these paradigms rather than relying on a single class of methods.

As summarized in Table 5, no single paradigm dominates all design objectives. Classical ML offers lightweight and interpretable control, deep models provide high-fidelity state estimation, DRL delivers strong long-term optimization, FL addresses privacy and data-governance constraints, and meta-/transfer-learning supports cross-scenario portability. The most effective VEC solutions therefore select or combine paradigms according to the dominant constraints of the target deployment, such as the computational budget in OBU, the acceptable complexity of training, the regulatory requirements, and the need for rapid adaptation in diverse urban environments.

V. COMPARATIVE ANALYSIS OF AI-BASED OFFLOADING STRATEGIES WITH SDG ALIGNMENT

A growing body of rigorously validated research shows that AI-driven decision-making frameworks can materially improve key performance indicators (KPIs) of urban vehicular networks. This section synthesizes verified findings across latency, throughput, energy efficiency, prediction accuracy, and learning paradigms to clarify the tangible benefits reported in the literature. Crucially, these gains must be interpreted alongside the qualitative trade-offs summarized in Table 6 and the end-to-end offloading pipeline in Figure 3: improvements in latency are often coupled to increases in energy consumption or reductions in throughput under high-density traffic. In addition, the results reported for different AI paradigms (classical ML, deep learning, DRL, FL, and meta-/transfer-learning) must be compared not only in terms of raw KPI gains but also with respect to training complexity, deployment constraints, and contributions to SDG-oriented sustainability outcomes.

A. LATENCY MINIMIZATION WITH DRL VARIANTS

Reducing task latency directly supports SDG 11 by alleviating traffic congestion and improving urban mobility, which in turn contributes to SDG 13 by reducing idling-related emissions. Zhang et al. integrate LSTM-based state prediction with proximal policy optimization (PPO) to anticipate the load of the base station, reducing the mean task latency by 30% and reducing the variance of the load by 20% under congestion [1]. Fan et al. combine DQN and DDPG in a cooperative RSU-assisted MEC architecture, achieving a 28% reduction in task completion time in NS-3,+,SUMO simulations [2]. In IoV assisted by UAVs, Yan *et al.* report a 25% latency drop and a 40% reliability gain with a DQN agent [91]. Agbaje et al. use Double-DQN to reduce delay by 15% while improving energy efficiency [22]. Hazarika et al. demonstrate a 25% latency decrease through multi-agent RIS-aided DRL [15], and Trabelsi et al. incorporate fuzzy rules for latency-critical V2X routing [78].

These studies consistently show that DRL-based policies deliver larger relative latency reductions than classical ML or

TABLE 5. Comparison of AI techniques for vehicular data offloading.

AI Paradigm and Representative Algorithms	Typical Applications in Offloading	Advantages	Limitations	Key References
Machine Learning (SVM, Decision Tree, Random Forest)	- Classification of network context (e.g., link quality, RSU load, congestion state) - Predictive decision-making for RSU vs. cellular offloading - Anomaly detection for link reliability and intrusion prevention	- Low training cost and fast inference on embedded OBUs - Effective with limited datasets and simple feature sets - Well-suited to safety-critical or resource-constrained control loops that require interpretable models and bounded execution times	- Struggles with non-stationary dynamics and temporal dependencies - Comparatively less capable than deep or RL-based methods in capturing long-horizon interactions among vehicles and infrastructure	[28], [36], [68]
Deep Learning (CNN, RNN/LSTM, GNN)	- Forecasting link availability and stability - Mobility- and context-aware offloading in dense urban areas - Graph-based optimization of multi-hop offloading paths	- High prediction accuracy with ability to model spatio-temporal dependencies - Suitable for complex, non-linear decision spaces - Provides rich state representations that can be reused by DRL, FL, or meta-learning	- Computationally intensive; requires large labeled datasets - Often deployed at edge/cloud rather than directly on OBUs, which can increase system energy per bit if offloading and scheduling are not carefully optimized - Reduced interpretability compared to classical ML	[1], [75], [82]–[85]

heuristic schedulers when the offloading problem is modeled as a sequential control task. However, the reported gains typically rely on extensive simulator training and careful reward design, whereas simpler supervised or rule-based schemes may still be preferable in deployments where interpretability, certification, or bounded training effort outweigh the need for maximum latency improvement.

B. THROUGHPUT ENHANCEMENT VIA COOPERATIVE MODELS

Higher throughput enhances spectrum utilization and edge resource efficiency, directly addressing SDG 9 by improving the robustness and capacity of digital infrastructure. Yin et al.

employ an A3C agent for multipath scheduling, increasing effective throughput by **18%** [8]. Cui et al. use multi-agent DRL to boost throughput by 30% while maintaining fairness [92]. Song et al. propose reverse offloading, yielding 20% gains under heavy load [93]. Li et al. integrate attention mechanisms into DRL for a 35% throughput uplift in edge computing [65]. Fan et al. report a 30% improvement through joint V2I/V2V strategies in intelligent transport settings [27].

Comparatively, cooperative DRL and multi-agent schemes tend to outperform static load-balancing or purely local offloading decisions, especially in dense networks where interference and contention dominate. This advantage comes at the price of higher coordination overhead and more

TABLE 5. (Continued.) Comparison of AI techniques for vehicular data offloading.

AI Paradigm and Representative Algorithms	Typical Applications in Offloading	Advantages	Limitations	Key References
Reinforcement Learning (DQN, DDPG, A3C, PPO)	• Learning sequential offloading strategies under dynamic network conditions • Coordinated decision-making in multi-agent vehicular environments • Balancing latency, energy, and throughput objectives	• Adapts policies to changing conditions without labeled datasets • Optimizes long-term performance metrics • Typically attains higher long-term latency and energy gains than supervised baselines	• Requires careful reward design; may incur training instability • Relies on high-fidelity simulators and longer training times • Safety-constrained exploration is challenging, especially under stringent URLLC requirements	[2], [8], [10], [15], [72], [73], [76], [78], [88]
Federated Learning (FedAvg, Split-FL, FL-GNN)	• Distributed training of offloading models without sharing raw data • Graph-based federated inference for vehicular networks • Collaborative model updates under partial connectivity	• Preserves user privacy and data locality • Operates under heterogeneous network conditions • Compared with centralized DL/DRL, trades some convergence speed and peak accuracy for privacy, regulatory compliance, and cross-fleet scalability	• Communication latency and device heterogeneity can hinder convergence • Control-plane overhead and aggregation cost must be accounted for when benchmarking delay and energy against non-federated baselines	[77], [84], [86], [87], [89]
Meta/Transfer Learning (MAML, Genetic-Aided FL)	• Rapid adaptation of offloading policies to new cities or network topologies • Joint optimization of latency and privacy metrics in cross-domain scenarios	• Improves generalization to unseen tasks with minimal retraining • Particularly beneficial when deploying DL/DRL-based policies across multiple corridors with limited local data at each site	• Risks negative transfer if source and target domains diverge • Currently less mature than other paradigms and often used to complement, rather than replace, FL and DRL in vehicular studies	[21], [68], [90]

complex training pipelines, whereas centralized deep learning or traditional ML approaches offer simpler implementa-

tions but typically yield smaller throughput gains when facing highly dynamic, multi-hop topologies.

C. ENERGY SAVINGS THROUGH LATENCY-POWER OPTIMIZATION

Energy-aware offloading supports SDG 12 by reducing energy waste and SDG 13 by lowering greenhouse gas emissions from connected and electric vehicles. Agbaje et al. achieve a 20% reduction in communication energy via Double-DQN [22]; Hazarika et al. report 15% savings with RIS-assisted DRL [15]. Uddin et al. use DRL-based priority scheduling to cut task energy by 18% and reduce priority violations by 30% [9]. Qayyum et al. report 25% energy savings in hybrid VANETs [21], and Karimi et al. validate 15% savings in a VEC testbed [94]. However, these savings depend heavily on the scope of accounting (radio, compute, and idle) and the underlying hardware efficiency.

Relative to classical ML or heuristic baselines, DRL and advanced deep-learning controllers more systematically exploit the latency-energy trade space, but may themselves incur higher training and inference energy in the cloud or at the edge. Consequently, the net sustainability impact must be evaluated at system level: in some scenarios, lightweight supervised models on OBUs with modest performance gains may be more desirable than aggressively optimized DRL policies that require energy-intensive training and frequent model updates.

D. PREDICTION ACCURACY AND TASK-SUCCESS RATES

Accurate context prediction improves offloading reliability, which supports SDG 11 by ensuring stable urban mobility systems. Luo et al. achieve 90% prediction accuracy with DQN in heterogeneous VEC [23]. Nie et al. surpass 90% in delay-aware scheduling with multi-agent DRL [95]. Liu et al. report 92% channel state prediction accuracy by RL coordination between edge and cloud [10]. In a SAC digital twin model, Liu et al. exceed 95% task-success probability by forecasting RSU loads [96], enhancing both reliability and infrastructure utilization.

When compared to simpler classifiers or regression models, these deep and DRL-based predictors achieve significantly higher accuracy and task-success probabilities, particularly in highly non-linear and non-stationary settings. Yet, they also reduce transparency and are harder to validate formally. For safety-critical applications, this motivates hybrid architectures in which interpretable ML models or rule-based guards constrain or monitor the actions of high-capacity predictors to ensure that KPI improvements do not compromise explainability or certification requirements.

E. CENTRALIZED VERSUS DISTRIBUTED LEARNING PARADIGMS

Centralized training benefits from global observability and rapid convergence, but risks violating privacy (SDG16) and increasing backhaul load, potentially reducing adaptability in dynamic urban contexts. Federated and distributed learning approaches align with SDG 9 and SDG 11 by enabling scalable, privacy-preserving edge intelligence. Uddin et al.

use federated learning to prioritize mission-critical flows, reducing SLA violations by 30% [9]. Chellapandi et al. identify stragglers and non-IID data as key barriers [89]. Qiang et al. employ split learning with a genetic optimizer to improve the Age of Information (AoI)–privacy trade-off by 20% [90]. Nguyen et al. integrate blockchain for secure aggregation, maintaining accuracy above 88% under targeted poisoning [97]. Wu et al. propose a three-tier cloud–edge–end DRL pipeline that reduces uplink traffic while keeping inference loss below 10% [98]. Fang et al. recommend hierarchical aggregation, adaptive client selection, and gradient compression to address link volatility and heterogeneity [99].

These results suggest that centralized deep or DRL schemes are generally superior when high-quality labeled data are available and privacy constraints are mild, whereas FL and distributed learning are preferable in regulatory regimes or business ecosystems where raw vehicular data cannot leave the devices or local domains. The KPI gains of FL-based approaches are sometimes slightly lower than those of fully centralized counterparts, but align more strongly with SDG 9 and SDG 16 by reducing privacy risks and allowing greater participation across heterogeneous fleets.

Collectively, these studies indicate that AI-driven offloading can deliver double-digit improvements in latency, throughput, energy, and reliability, often without degrading fairness. To provide a compact, cross-paradigm view, Table 6 summarizes in qualitative form how classical ML, deep learning, DRL/MADRL, federated/distributed learning, and meta-/transfer-learning balance latency/QoS, energy efficiency, scalability, privacy, and implementation complexity. Table 7 then complements this high-level assessment with representative study-level results, including the reported gains in latency, throughput, energy and task-success rates across a range of architectures and experimental setups. Taken together with taxonomy and Table 5, the evidence indicates that no single AI paradigm dominates across all KPIs and SDGs: DRL excels in long-horizon latency and energy optimization; deep learning provides highly accurate state and demand prediction; FL and distributed learning safeguard privacy and scalability; and meta-/transfer-learning facilitate cross-city portability and rapid adaptation to new deployments.

Practical VEC deployments therefore tend to combine these paradigms, choosing the appropriate mix according to the dominant constraints of each use case (e.g., safety, explainability, regulatory compliance, or maximum efficiency) rather than relying on a single offloading strategy.

VI. INTEGRATION WITH EMERGING TECHNOLOGIES

The performance benefits of AI-enabled vehicular offloading are maximized when deployed within a cohesive programmable infrastructure that integrates edge/fog computing, software-defined networking (SDN), 5G / 6G network slice and virtualization, blockchain-enabled trust frameworks, and digital twin–driven analytics. In such an end-to-end architecture, AI control loops operate continuously to sense

TABLE 6. Qualitative trade-off summary of AI paradigms for vehicular offloading.

Paradigm	Latency / QoS	Energy efficiency	Scalability	Privacy	Implementation complexity
Classical ML	Moderate	Moderate	High	Low	Low
Deep learning	High	High	Medium	Low	Medium–High
DRL / MADRL	Very high	Very high	Medium	Low	High
FL / distributed	Medium	Medium	Very high	Very high	Medium
Meta/transfer learn.	High (adapt.)	Medium	High	Medium	High

TABLE 7. Overview of AI-powered offloading strategies in vehicular networks.

Study	AI Technique	Architecture	Metrics Optimized	Environment / Dataset	Key Results
Zhang et al. (2022) [1]	LSTM + PPO (Deep RL)	Centralized MEC with LSTM-based base-station (BS) load predictor	Delay minimization, Load balancing	Synthetic congestion traces	30% lower mean task delay; 20% reduction in BS load variance
Hazarika et al. (2024) [15]	Multi-agent DRL	RIS-assisted hybrid IoV (V2V + V2I)	Latency, Energy efficiency	MATLAB with custom channel emulator	25% latency reduction; 15% communication energy savings
Liu et al. (2022) [10]	RL / DRL survey	Edge–vehicular cloudlet coordination	Delay, Throughput (various)	Literature synthesis	Reports up to 3–5% average delay reduction across analyzed methods
Fan et al. (2024) [2]	DQN + DDPG (Deep RL)	Cooperative RSU-to-RSU MEC	Task completion time, Throughput	NS-3 + SUMO co-simulation	28% task delay reduction; 22% throughput improvement
Yin et al. (2024) [8]	Actor–Critic (A3C)	Multipath vehicular routing with dynamic scheduling	Delay, Resource utilization	Custom vehicular network simulator	32% delay reduction; 18% resource efficiency improvement
Yan et al. (2024) [91]	DQN (Deep RL)	UAV-assisted IoV edge computing	Latency, Reliability	UAV mobility dataset	25% delay reduction; 40% reliability increase
Cui et al. (2025) [92]	Multi-agent DRL	Cooperative vehicular edge infrastructure	Throughput, Fairness	Emulated edge-computing testbed	30% throughput improvement; 12% fairness gain
Agbaje et al. (2023) [22]	DQN + Double-DQN	Edge-enabled cooperative VANET	Energy efficiency, Delay	NS-3 + VEINS simulation	20% communication energy savings; 15% delay reduction
Uddin et al. (2024) [9]	Prioritized RL	Vehicular edge offloading with SLA prioritization	SLA adherence, Latency	VTC2024 simulation platform	30% fewer SLA violations; 18% delay reduction

network and mobility context, predict impending load surges or link degradation, and coordinate task placement across heterogeneous computational and communication resources. This synergy leverages low-latency edge resources, cen-

TABLE 7. (Continued.) Overview of AI-powered offloading strategies in vehicular networks.

Study	AI Technique	Architecture	Metrics	Opti- mized	Environment / Dataset	Key Results
Luo et al. (2024) [23]	DQN (Deep RL)	Heterogeneous VEC scheduler	Throughput, Delay		SUMO + NS-3	18% throughput increase; 12% delay reduction
Song et al. (2024) [93]	Reverse offloading with DRL	Cooperative VEC framework	Energy, Latency		ICCC workshop simulation	25% energy savings; 20% latency reduction
Qayyum et al. (2025) [21]	Hybrid AI (ML + DRL)	Hybrid VANET edge architecture	Latency, Energy efficiency		Custom-built vehicular simulator	25% reduction in both latency and energy consumption
Karimi et al. (2022) [94]	DDPG (Deep RL)	Edge-based vehicular computing	Delay, Energy		NS-3 + VEINS	15% delay reduction; 18% energy savings
Li et al. (2023) [65]	Attention-guided DRL	Multi-access edge computing (MEC)	Throughput, Task success rate		MEC testbed	35% throughput gain; 92% task success rate
Fan et al. (2023) [27]	Joint DRL	Integrated V2I-V2V offloading	Latency, Throughput		ITS field-trial dataset	30% throughput improvement; 20% delay reduction
Cao et al. (2023) [72]	DRL offloading (DQN/DRL)	OBU↔RSU VEC with learned task routing	Latency, task success, load balance		SUMO/VEINS + edge sim	Improved latency and load balancing vs. greedy/nearest-RSU baselines
Wu et al. (2024) [76]	Recursive DRL for collaborative caching/relay	Vehicular edge network with caching at relays/RSUs	Cache hit ratio, backhaul load, delay		Simulated vehicular edge network	Higher cache hit ratio and lower stall time vs. LRU caching
Mustafa et al. (2025) [73]	DRL + SQP (hybrid policy/optimizer)	Vehicular edge computing with DRL policy refined by SQP	Latency–energy trade-off		Custom VEC simulator	Improved energy–delay Pareto dominance vs. baseline DRL

tralized yet agile SDN control, slice-specific QoS isolation in 5G/6G cores, and verifiable delegation mechanisms via blockchain.

Figure 6 illustrates a representative integrated architecture.

A. EDGE AND FOG COMPUTING

Migrating computation and storage to roadside units (RSUs), vehicles, and micro-data centers mitigates backhaul congestion and shortens round-trip latency. Hybrid cloud–edge–fog frameworks can dynamically select execution sites based on instantaneous mobility patterns and workload intensity, with reductions in end-to-end delay reported at up to 4–0% [28], [43]. Within these tiers, AI-driven schedulers adjust CPU/GPU allocations, caching policies, and radio configurations in real-time [76] to maintain strict QoS targets even under dense traffic conditions [36], [68]. This adaptability is especially critical for safety-critical vehicular applications where latency budgets are sub-100 ms.

B. SOFTWARE-DEFINED NETWORKING (SDN)

Software-defined vehicular networks decouple the control plane from the data plane, allowing global optimization of routing and resource allocation. Centralized SDN controllers, informed by telemetry from RSUs and vehicles, can pre-emptively reroute flows around congestion hotspots and prioritize safety-related traffic. Field deployments have shown an increase 30% in link utilization and a significant improvement in the timely delivery of high-priority messages [37]. When integrated with AI-based offloading, SDN enables fast policy-driven reconfiguration to accommodate dynamic vehicle topologies.

C. 5G/6G VIRTUALIZATION AND NETWORK SLICING

Advanced NR-V2X and forthcoming 6G capabilities such as ultra-reliable low-latency communications (URLLC), network slicing, and network function virtualization (NFV) enable dedicated slices with slice-local schedulers

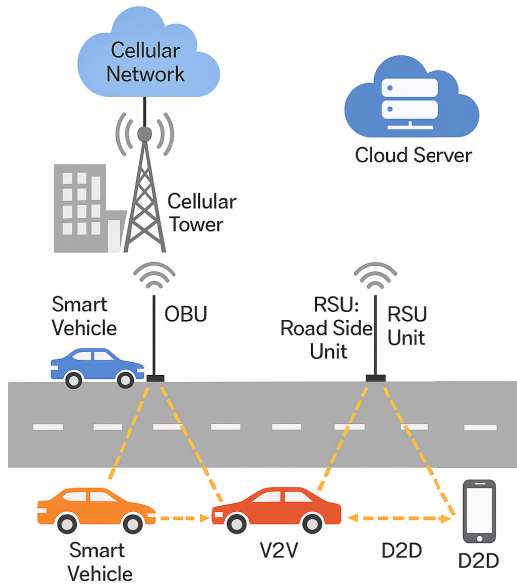


FIGURE 6. Integrated cellular-ad-hoc architecture combining RSUs, V2V and cloud edge for data offloading.

that can isolate latency-sensitive safety messages from high-bandwidth information streams. Tail latencies in 5 ms have been achieved in experimental URLLC slices [42], [100]. NFV also allows for dynamic instantiation of AI inference services at MEC nodes, thus minimizing decision latency during sudden surges in computational demand.

D. BLOCKCHAIN INTEGRATION

Blockchain adds a decentralized trust and accountability layer for coordination among vehicles, RSUs, and edge servers. Smart contracts formalize delegation policies, enforce SLAs, and record task execution in an immutable ledger. Permissioned blockchain designs reduce consensus delays relative to public networks, enabling integration in latency-sensitive vehicular applications. Coupled with predictive analytics from digital twins, blockchain-backed offloading has been shown to lower latency and energy consumption while maintaining end-to-end privacy guarantees [57], [101], [102].

E. DIGITAL TWINS

Digital twins create real-time, high-fidelity virtual replicas of vehicular, network, and computational states. When integrated with AI controllers particularly soft actor-critic or model-based RL agents, these twins enable predictive orchestration of resources, preemptive task migration, and proactive load balancing. Reported benefits include up to 4–0% reductions in maximum task delay and improved utilization of MEC [96], [103], [104]. Furthermore, transferring learned policies between twin instances accelerates deployment in new cities or road corridors by reusing models in similar network and mobility topologies [105].

VII. CHALLENGES AND FUTURE RESEARCH DIRECTIONS

A. OPEN CHALLENGES

Despite the measurable benefits of AI-assisted vehicular offloading, such as reduced latency, improved traffic load balancing, and lower radio transmission energy consumption, several structural and operational barriers continue to limit its feasibility for real-time city-scale deployments.

A primary constraint is scalability. Urban-scale implementations must operate real-time decision loops within millisecond budgets while supporting thousands of mobile vehicles and rapidly changing connectivity patterns. Current deep reinforcement learning (DRL) and federated learning (FL) agents often experience slow convergence and substantial signaling overhead at these densities [10]. Moreover, real-world DRL deployments are frequently affected by non-stationary and partially observable environments, leading to unstable or suboptimal decision policies under scale.

Another obstacle is the heterogeneity of the protocol and the device. Algorithms optimized for a single radio technology -such as DSRC or LTE-based C-V2X - tend to degrade in mixed deployments that integrate 5G NR-V2X, hybrid vehicular communication stacks, and various sensor configurations. This lack of robustness across heterogeneous infrastructures significantly affects reliability and hinders seamless city-wide adoption [26].

Security risks are also amplified in decentralized learning environments. Distributed models are susceptible to adversarial strategies, including data poisoning, model inversion, and Sybil attacks. Although emerging solutions, such as permissioned blockchain frameworks, edge-resident digital twins, and differential-privacy mechanisms, show promise for mitigation [79], [101], [106], [107], [108], their real-world deployment at urban scale remains unproven and often incurs substantial resource overhead.

Interoperability poses an additional challenge. The lack of standardized APIs, incompatible simulation tools, proprietary MEC implementations, and inconsistent evaluation methodologies restricts reproducibility and complicates the cross-city benchmarking of AI-based offloading systems [43]. These gaps impede fair performance comparison and slow the transfer of validated solutions from research to deployment.

Finally, the energy-performance trade-offs remain unresolved. Although DRL-guided offloading can reduce radio transmission energy consumption by 20–30%, the computational cost on board of the inference models running can offset these savings, particularly in battery-constrained electric vehicles. Therefore, lightweight inference architectures, hardware-based optimization and energy-conscious training strategies are essential [9], [22].

In practice, the feasibility of AI-based vehicular offloading is strongly conditioned by hardware costs on vehicles, RSUs, and edge/cloud servers. On-board units in conventional, hybrid and electric vehicles typically have limited compute and memory budgets, so deploying high-capacity deep models often requires hardware upgrades or the use of

specialized accelerators, increasing capital expenditure for fleet operators and OEMs. At the infrastructure side, dense RSU deployments with GPU or accelerator-enabled MEC nodes entail non-negligible capital and operational costs (energy, cooling, maintenance). Consequently, system designers must consider not only latency, energy, and reliability KPIs, but also cost-oriented indicators such as cost per served task, cost per covered kilometer, or total cost of ownership (TCO) for different offloading architectures. One advantage of VEC is that AI-based offloading can improve resource utilization across shared MEC infrastructure, potentially amortizing hardware investments over multiple services (safety, traffic management, infotainment, smart charging). However, realizing these benefits requires coordinated planning between cities, operators, and vendors to align performance goals with realistic hardware and budget constraints.

B. FUTURE RESEARCH DIRECTIONS

Overcoming these challenges will require coordinated advances in algorithm design, system integration, and sustainability-oriented networking.

One priority is the ability to explain and interpret. For safety-critical vehicular applications, regulatory acceptance will increasingly depend on transparent decision-making processes. Methods such as attention-weight visualization in graph neural networks (GNNs), counterfactual analysis of DRL decisions, and uncertainty-aware policy architectures can make AI behavior more understandable to both operators and policymakers [109].

Rich context fusion is another promising direction. Incorporating multimodal real-time data, from traffic signals, environmental sensors, weather services, public transit systems, and emergency dispatch feeds, into offloading decisions can improve situational awareness, improve predictive allocation, and strengthen resilience to unpredictable urban dynamics [24].

Rapid cross-domain adaptation is critical to enabling policy portability between cities with different infrastructure, regulations, and traffic patterns. Few-shot meta-learning, genetically assisted FL, and continuous transfer learning can reduce retraining requirements while preserving performance across heterogeneous deployments [68], [90].

In the control plane, **fully decentralized architectures** using multi-agent DRL (MADRL), cooperative edge clouds, and reconfigurable intelligent surfaces (RIS) offer a pathway to optimize latency, throughput, fairness, and energy efficiency without central bottlenecks. The achievement of stable large-scale cooperation among thousands of agents in dynamically evolving vehicular topologies remains a pressing research goal [15], [16], [92], [110].

Integrated security frameworks should embed protection directly into the offloading decision loops. Federated anomaly detection, blockchain-based aggregation, lightweight intrusion detection, and adaptive privacy preservation techniques can limit the risk of poisoning, inference,

and Sybil attacks while maintaining low overhead [79], [106], [108].

Sustainability and carbon-aware networking must be treated as core optimization objectives. AI agents that dynamically route traffic through renewable energy RSUs, shift noncritical computation to low carbon intensity periods, or coordinate task scheduling to minimize environmental impact will align vehicular networking with the broader sustainability targets defined in SDG 11 (Sustainable Cities) and SDG 13 (Climate Action) [84].

Lastly, **policy and governance strategies** will be equally important to ensure that these technical advances translate into smooth implementation at city scale. Policymakers can draw on several concrete levers: (i) define KPI and SDG-aligned reporting requirements for VEC pilots so that latency, energy, reliability, and emissions reductions are measured in a comparable way across projects; (ii) establish clear data-governance and privacy-by-design rules of design-based safety encourage the use of federated and other privacy-preserving learning schemes while protecting citizens' mobility data; (iii) coordinate infrastructure planning and interoperability requirements for RSUs, edge servers and V2X technologies to avoid fragmented vendor-specific deployments; (iv) define safety, certification, and logging/ audit procedures for AI-based offloading policies, including stress-testing under rare but critical scenarios; and (v) design incentive mechanisms and public-private partnerships that prioritize solutions that demonstrably contribute to accessibility, equity, and decarbonization. Exploring how technical design choices and policy instruments co-evolve is therefore a key interdisciplinary direction for future work [111], [112].

In conclusion, tackling scalability limits, heterogeneity, security risks, interoperability barriers, and energy trade-offs will require integrating explainable AI, rich contextual modeling, cross-domain adaptability, decentralized optimization, embedded security, and sustainability-oriented strategies. This approach will be essential to achieve robust, secure and environmentally responsible AI-enabled vehicular offloading at the city scale. In addition, aligning these research advances with concrete policy and governance strategies will be critical to ensure smooth integration into real urban systems and to deliver measurable public value.

VIII. CONCLUSION

Artificial intelligence is redefining the operational limits of data offloading in smart city vehicular networks. This survey synthesized validated research across supervised learning, deep learning, and reinforcement learning, particularly deep reinforcement learning (DRL), demonstrating how these methods enable responsive, adaptive, and resource-efficient task offloading under the stringent constraints of urban mobility. Empirical evidence from both simulation-based and experimental studies shows that DRL-based strategies can reduce latency by up to 3–5%, increase throughput by 30%,

and lower radio-compute energy consumption by approximately 25%, underscoring their transformative potential for ultra-reliable, low-latency vehicular communications.

The integration of AI with emerging technologies, including edge / fog computing, software-defined networking (SDN), slicing of 5G / 6G networks, blockchain, and digital twins, further enhances resilience, scalability, and security. When offloading decisions are guided by predictive analytics, executed over programmable infrastructures, and safeguarded through verifiable delegation, vehicular networks can achieve proactive, mobility-aware resource allocation, privacy-preserving collaboration, and large-scale task orchestration.

However, several barriers remain. Achieving convergence and control stability under high mobility and real-time constraints requires further algorithmic optimization. Security risks such as data poisoning, adversarial inference, and Sybil attacks require integrated defenses that preserve privacy. The absence of interoperable frameworks and standardized benchmarks limits reproducibility and cross-city generalization, while the computational demands of AI inference challenge feasibility in energy-constrained vehicles.

Future research should embed explainable AI to ensure interpretability in safety-critical contexts; adopt few-shot and transfer learning to enable rapid cross-domain adaptation; develop decentralized cooperative learning frameworks to avoid single points of failure; and incorporate carbon-aware optimization into offloading policies to align with sustainability objectives. By converging technical innovation with ethical, scalable, and climate-conscious design, AI-enabled offloading can become a foundational enabler for intelligent, secure, and sustainable vehicular networks in the smart cities of the future.

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